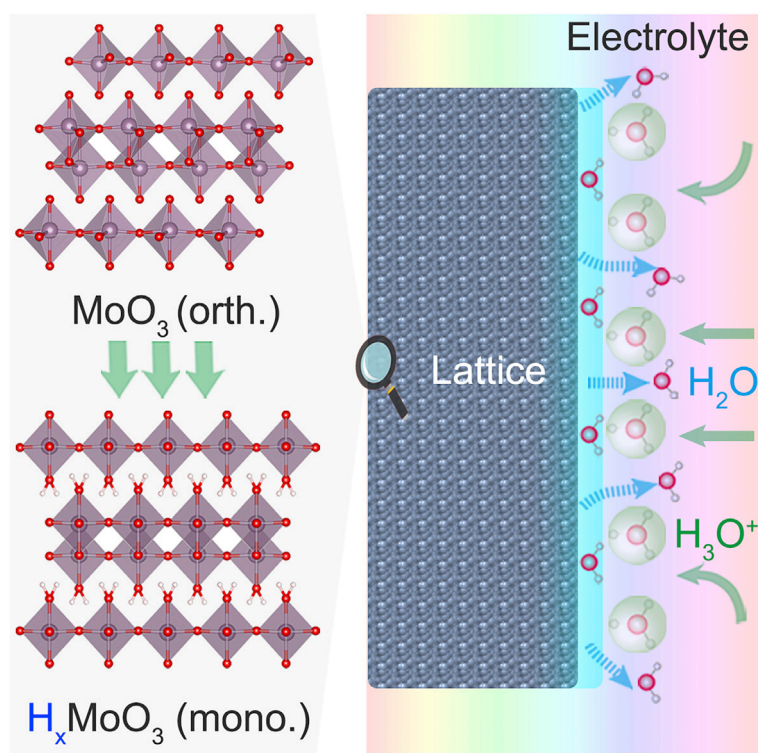


Article

Two-Phase Electrochemical Proton Transport and Storage in α -MoO₃ for Proton Batteries



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chuan.zhao@unsw.edu.au

HIGHLIGHTS

Operando XRD and EQCM reveal two scenarios of proton transport and intercalation

Hydroniums transport charge at interfaces with inverse water movement

Naked protons insert into electrode lattices to trigger bulk structure transition

Water activities induce surface hydration and influence cycling stability

Hydrogen ions (proton/hydronium) are promising charge carriers for future high rate and capacity energy storage. Here, Guo et al. investigate the protonation topochemistry of α -MoO₃ involving sophisticated hydronium/water interplay with electrode surfaces and proton insertion-triggered bulk reactions, which enable a diffusion-dominated electrode also to present rapid rates.

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Article

Two-Phase Electrochemical Proton Transport and Storage in α -MoO₃ for Proton Batteries

Haocheng Guo,¹ Damian Goonetilleke,^{1,3} Neeraj Sharma,¹ Wenhao Ren,¹ Zhen Su,¹ Aditya Rawal,² and Chuan Zhao^{1,4,*}

SUMMARY

Diffusion-controlled charge storage and phase transitions of electrodes are typical indicators of sluggish kinetics in battery chemistries. However, fast rate capabilities are found in an α -MoO₃ proton intercalation electrode that presents both features. Here, the unique topochemistry is shown to involve multiple ion-electrode interactions and proceeds via two key steps: hydronium adsorption on surfaces and proton insertion into bulk lattices. This triggers structure transitions from MoO₃ to hydrogen molybdenum bronzes (HMBs). Following the first process, subsequent rearrangements proceed only among HMBs phases with high reversibility and kinetics, thus providing structural explanations to the fast rate capability. At electrode-electrolyte interfaces, hydronium is the active charge carrier that initiates charge transfer and surface hydration, which are accompanied by water adsorption/desorption with reduced polarization and enhanced kinetics. Further water activity is shown to induce material dissolution during function. These findings offer fundamental insights in proton chemistries, which may form the basis of future high rate and capacity energy storage.

INTRODUCTION

Batteries and supercapacitors are representative energy storage techniques that present high capacity (batteries) and high rate capability and long cycle life characteristics (supercapacitors). To develop devices integrating both of their advantages is potentially revolutionary, yet very challenging.¹ This is because of the intrinsic conflicting origins of capacity and rate capability: high capacity requires more redox reactions and longer diffusion distance of ion charge carriers, but the corresponding kinetics typically become slow. The discovery of intercalation pseudocapacitance demonstrates that the fast capacitive charge storage can no longer be surface limited,² therefore somewhat bridge rate and capacity from the material perspective. Otherwise, the choice of ion charge carriers also plays a crucial role in rate capabilities as ion transport contributes to the overall kinetics.³ Rechargeable batteries are dominated by metal cations as charge carriers, predominantly Li⁺,⁴ then systems of Na⁺,⁵ K⁺,⁶ and Mg²⁺,⁷ Al³⁺,⁸ Zn²⁺,⁹ and Ca²⁺¹⁰ have also been investigated as cost-effective alternatives and/or to extend the knowledge of battery chemistry. Although these feature various advantages, their kinetics essentially rely on long-distance diffusion of metal cations with increased weights and sizes; therefore, it is hard to achieve a breakthrough in rate capability.

Protons are promising charge carriers that can overcome these hurdles.³ Proton is the lightest of all ions and presents the smallest ionic radius, which is favorable for ion

¹School of Chemistry, Faculty of Science, University of New South Wales, Sydney, NSW 2052, Australia

²Mark Wainwright Analytical Centre, University of New South Wales, Sydney, NSW 2052, Australia

³Current address: Battery and Electrochemistry Laboratory (BELLA), Institute of Nanotechnology, Karlsruhe Institute of Technology, Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany

⁴Lead Contact

*Correspondence: chuan.zhao@unsw.edu.au
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transport. It can be loaded as a polyatomic ion and move diffusively, such as H_3O^+ , NH_4^+ , and so forth (vehicle mechanism¹¹). Proton can also be transported via the Grotthuss mechanism,¹² a unique hopping via the translational events of naked protons through concerted cleavage and formation of both the O-H covalent bonds and the $\text{H}^+\cdots\text{OH}_2$ hydrogen bonds. This conduction can be extremely fast in aqueous electrolytes and even in the solid state.³ Furthermore, hydrogen is among the most abundant elements on Earth and is therefore suitable for grid-scale application. All of these characteristics make proton/hydronium batteries a promising solution for future sustainable energy storage.¹³ Thus far, several materials have been found to be active in faradic proton/hydronium storage, including organic solids,^{14,15} pseudocapacitive metal oxides ($\text{RuO}_2 \cdot x\text{H}_2\text{O}$, MnO_2),^{16,17} MXenes,¹⁸ hydrogen insertion oxides,^{19,20} and Prussian blue analogs.^{3,21} Some of them have been adopted recently in electrochemical devices with decent performance.^{18,22–24} For all applications, the electrode function mechanisms are of vital importance, but the current understanding is limited.

Here, the proton transport and storage fundamentals are investigated via targeting $\alpha\text{-MoO}_3$. Unlike the (pseudo-)capacitive nature in most of the available materials to date, $\alpha\text{-MoO}_3$ presents diffusion-dominant charge storage with the highest known capacities (typically sluggish), but also fast rate capabilities, even in commercial materials with bulk sizes (Figure S1). Therefore, it can be ideal for understanding the potential of proton chemistries. Until now, studies have reported proton storage performances of $\alpha\text{-MoO}_3$,^{20,22,25} and suggest that it is promising for future high capacity and power devices. However, knowledge of the function mechanisms is still vague; in particular, there are key disputes/questions as to whether proton²⁰ or hydronium²² is the real charge carrier, which can restrict further developments. In this study, we disclose the topotactic proton (de)intercalation of $\alpha\text{-MoO}_3$ through a combination of *operando* synchrotron X-ray diffraction (XRD) and electrochemical quartz crystal microbalance (EQCM) techniques, which allow real-time monitoring of both structure and mass evolution. Further analytical methods are also applied to comprehensively understand the ion-electrode-interactions and other fundamental processes during operation.

RESULTS AND DISCUSSIONS

Electrochemistry of MoO_3 in Acidic Electrolytes

The basic electrochemistry of MoO_3 electrodes for proton storage was investigated using a three-electrode system with Ti-loaded electrodes in 4.4 M H_2SO_4 (unless otherwise noted), an acidic electrolyte commonly used in lead-acid batteries.¹⁴ A series of cyclic voltammograms (CVs) was recorded to study the redox process and kinetics of the MoO_3 electrodes. The Figure 1A inset shows a typical CV of MoO_3 obtained after the initial activation cycles. Three pairs of redox peaks are obtained at a low scan rate of 0.3 mV s^{-1} , designated as C1/A1, C2/A2, and C3/A3, respectively. These peaks correspond to hydronium adsorption/desorption and proton (de)intercalation processes, which will be discussed in detail later. As shown in Figure S2, hydrogen evolution/oxidation reactions (HER/HOR) occur at more negative potentials that are distinct from the C3/A3 redox; therefore, HER can be avoided by tuning the cutoff potential in our case (see also the discussion in Note S1). At higher sweep rates, the redox pair C1/A1 merges with C2/A2, forming a pair of broad peaks. With varying sweeping rates (v), the current (i) responds by a power-law function, $i = av^b$,^{26,27} where a and b are variable coefficients. As displayed in Figure 1B, the calculated b values are found to be 0.564/0.579 for C2/A2, and 0.508/0.559 for C3/A3. These numbers are close to the value expected from the semi-infinite diffusion-controlled kinetics ($b = 0.5$), thus suggesting that charge storage is diffusion dominant.

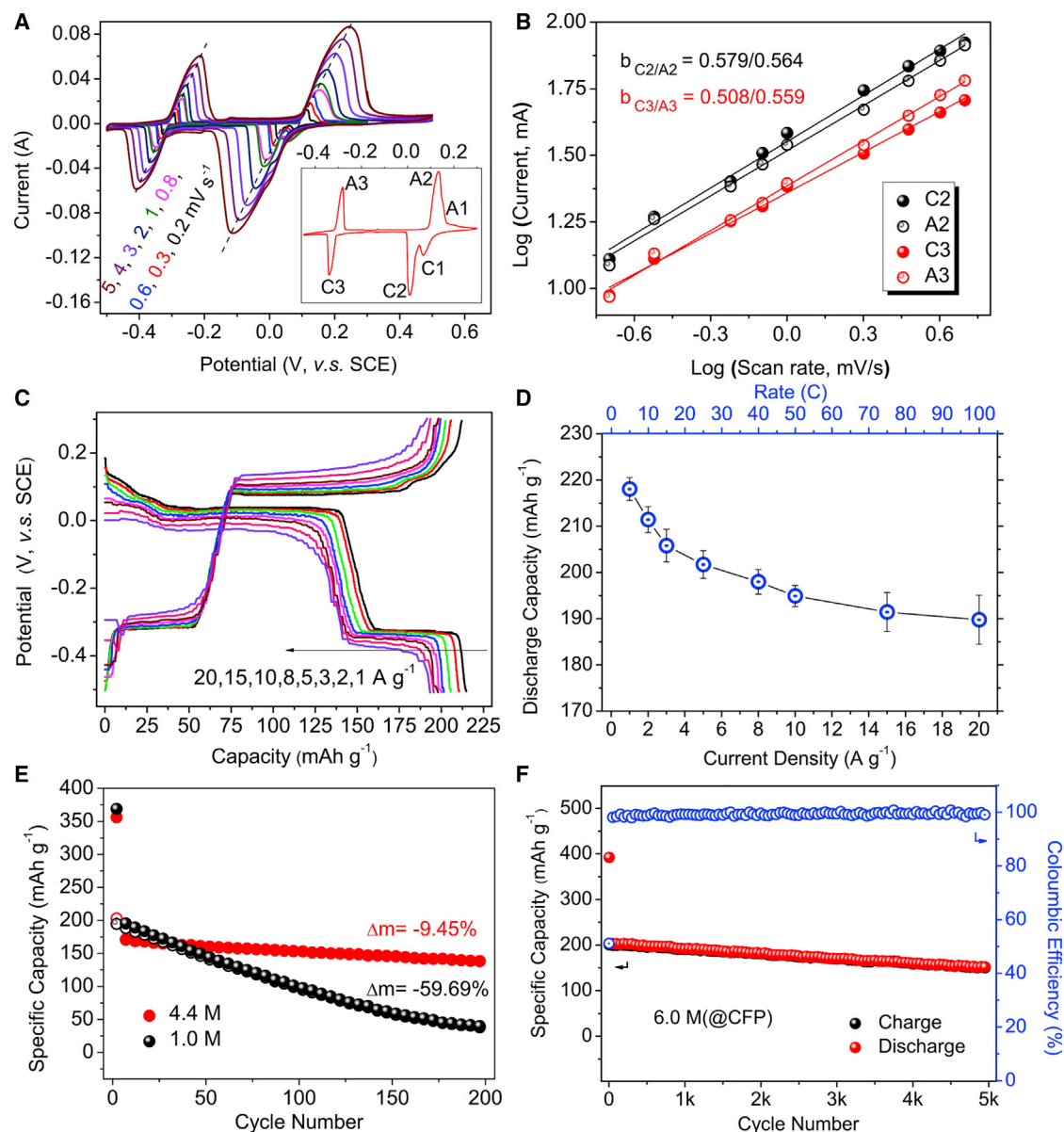


Figure 1. Electrochemical Properties of MoO₃ for Proton Storage

(A) Cyclic voltammograms (CVs) obtained at various scan rates in 4.4 M H₂SO₄; the inset presents the CV at the scan rate of 0.3 mV s⁻¹.

(B) Log values of peak currents versus scan rates of (A).

(C and D) The charge-discharge curves of MoO₃ anode under different current densities in 4.4 M H₂SO₄ (C), and (D) the corresponding plot of discharge capacity as a function of current density. Assuming 1 C = 200 mA g⁻¹, the rate performances of 5, 10, 15, 25, 40, 50, 75, and 100 C are displayed. The error bars indicate 1 SD.

(E) Cycling comparison of double-sided coated electrodes in 4.4 M and 1.0 M H₂SO₄ electrolyte (active material mass ~10 mg@Ti-plates, 1 A g⁻¹ for the initial 2 cycles, then 5 A g⁻¹ for the following cycles); Δm indicates post-cycled electrode mass changes.

(F) Cycling performances for MoO₃@CFP electrode in 6.0 M H₂SO₄ electrolyte at 10 A g⁻¹ throughout (see also Figure S6).

Figure 1C presents the reversible specific discharge-charge curves at different rates.

It is observed that the start of the discharge curve becomes flattened upon current increases and presents noticeable polarization. This correlates well with the disappearance of the A1/A2 processes mentioned above, and both are reflections of a relatively sluggish interface reaction (discussed later). Similar levels of polarization are observed for the other plateaus at higher currents and imply similar

electrochemical origins for these redox processes after activation (e.g., iR resistances). These similarities in polarization differ from the current-change-response (Figure S3), but are similar to the concentration-change response of the first cycle (Figure S4). Detailed relationships of the discharge capacity versus current density are shown in Figure 1D. At 1 A g^{-1} (assuming $1 \text{ C} = 200 \text{ mA g}^{-1}$, then it is 5 C), a high average value of $\sim 218 \pm 4 \text{ mAh g}^{-1}$ can be delivered. Our results from micron particles are close to those of recent reports.^{22,25} (Note, however, that the particle size, morphology, crystallinity, and so forth of MoO_3 may also influence performance.²⁰) With increasing current, the capacity decreases and stabilizes gradually. When reaching 20 A g^{-1} (100 C), $\sim 87\%$ of the 1 A g^{-1} capacity is retained, demonstrating excellent rate capabilities. Such superior rate performances suggests fast reaction kinetics and contrasts with other cation systems, where diffusion-controlled processes are typically sluggish.

When tested in different electrolyte concentrations (Figures 1E and S5), the electrodes display similar rate responses up to high currents ($>15 \text{ A g}^{-1}$), while different cycling performances are presented and the standard group ($4.4 \text{ M H}_2\text{SO}_4$) clearly surpasses the dilute contrast ($1.0 \text{ M H}_2\text{SO}_4$) in stability. Moreover, mass losses are also detected after cycling and consistent with the stability differences, as the $1.0 \text{ M H}_2\text{SO}_4$ group suffers an evident loss of $\sim 60\%$. This highlights a material dissolution issue during electrochemistry.²⁸ Figures 1F and S6 verify that cycling stability can be further enhanced in more concentrated electrolytes (e.g., $\sim 75\%$ capacity retention after 5,000 cycles in $6.0 \text{ M H}_2\text{SO}_4$). These mean instead of common cases of acidic corrosion (see also the morphology differences in Figure S7 and discussions in Note S2), the oxide dissolution herein is induced by sophisticated water activities detailed later. Note, however, that corrosion should still be considered for other cell components (Figures S8–S10); therefore, developing less corrosive protic electrolytes or alternative strategies to restrain water activities could be more promising for practical applications. Overall, these distinct behaviors are believed to result from the unique properties of proton storage and transportation in MoO_3 , which are disclosed step by step below.

Initial Proton (De)intercalation and Operando Structure Evolution

When reacting with hydrogen at ambient temperatures,^{29,30} MoO_3 forms a series of hydrogen molybdenum bronzes (HMBs, H_xMoO_3 , $0 < x < 2$), in which four phase ranges are identified.³¹ Figure 2A exhibits the charge-discharge profiles of the MoO_3 electrode in the first two cycles. In the first discharge, three major redox steps are observed, with the first small plateau accounting for $\sim 0.2 \text{ mol e}$, the second flat plateau for $\sim 1.3 \text{ mol e}$, and the third short flat plateau for $\sim 0.3 \text{ mol e}$. There are also slopes among these redox steps, which compensate for the remainder of the discharge. These plateaus and slopes correspond to the redox peaks C1', C2' (both irreversible), and C3 and the non-redox regions in the initial CV profiles discussed later. A high capacity of $\sim 382 \text{ mAh g}^{-1}$ is obtained at the end of the first discharge, which is equivalent to 2.05 mol H^+ intercalating into the electrode according to Faraday's law, and matches well with the theoretical maximum value of 372.4 mAh g^{-1} for H_2MoO_3 .

Upon charging, corresponding redox steps are also observed, but the profile is different. While the low-potential charge plateau (A3) shows good reversibility, the high-potential plateau (A2) is significantly shorter than that in the 1st discharge (C2') and exhibits a noticeable overpotential. After the A2 plateau, there is a very small bump before the slope to the potential cut-off that can be assigned to the small redox peak A1 in the CV (Figure 1A, inset). Overall, 219 mAh g^{-1} capacity

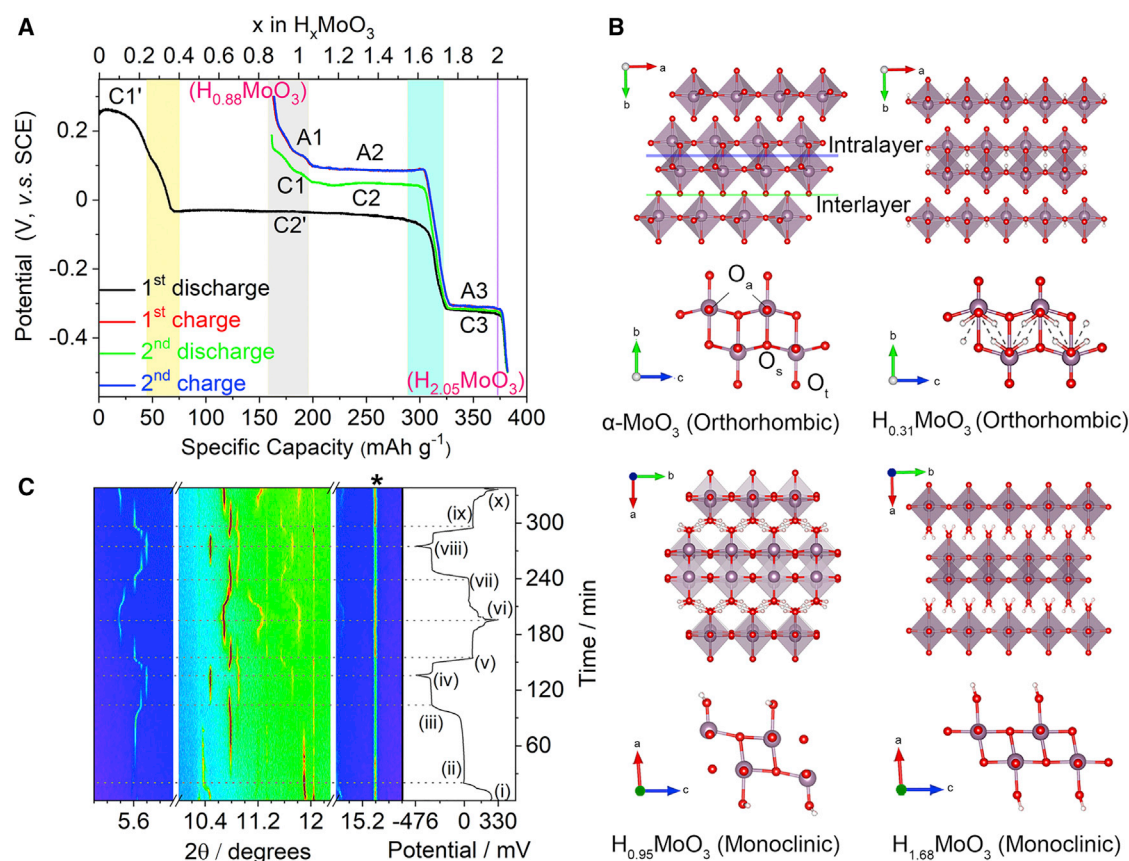


Figure 2. Electrochemistry and Operando Structure Changes of Initial Proton (De)intercalation

(A) The initial 2 galvanostatic charge-discharge profiles of MoO_3 at 1 A g^{-1} . Typical redox processes are labeled as C (cathodic)/A (anodic) n , where n is the redox index. Note the initial 2 cathodic processes are unique and labeled as $\text{C1'}/\text{C2'}$.
(B) Structure diagrams of $\alpha\text{-MoO}_3$ and the hydrogen molybdenum bronzes (HMBs) with different hydrogen amounts. The purple, red, and white spheres refer to Mo, O, and H atoms, respectively. Note that these structures were defined by different authors; therefore, the coordinates are different.
(C) The contour plot of operando diffraction data of cell 1 (in $4.4 \text{ M H}_2\text{SO}_4$ electrolyte, galvanostatically tested at $\sim 1 \text{ C}$), correlated with the electrode potential as a function of time. Different stages of the electrochemical reaction are labeled with Roman numerals. (The refinement results are shown in Table 1 and Figure S16) The asterisk indicates signal from Ti substrates.

(i.e., 1.18 mol proton extraction) is obtained, suggesting a significant capacity loss after the first cycle and a nominal product composition of $\text{H}_{0.88}\text{MoO}_3$. These changes in proton content can be verified by the chemical shift of Mo^{n+} that accounts for charge compensation (Figure S11). Interestingly, when tuning the current density ($1/6\text{--}10 \text{ A g}^{-1}$) or electrolyte concentration ($1.0\text{--}6.0 \text{ M H}_2\text{SO}_4$), the capacities obtained are similar despite the different polarizations (see Figures S3 and S4). This suggests that the amount of proton exchange is not limited by condition variation within these ranges. After the 1st activation cycle, the electrode was found to discharge and charge reversibly, with symmetric redox behaviors and reversible capacities of $\sim 219 \text{ mAh g}^{-1}$. These results highlight that understanding the initial reactions is critical to optimize the performance of MoO_3 and to achieve increased reversibility.

The initial electrochemical curves also suggest a possible reaction path for hydrogen uptake in MoO_3 , with representative structures illustrated in Figure 2B. MoO_3 typically has an orthorhombic layered structure of edge-sharing $[\text{MoO}_6]$ octahedra linked via vertices, and the bilayer frameworks are stacked together via van der

Waals (vdW) force. The structure contains 3 oxygens of interest in the lattice—the “terminal oxygen” (O_t), the “asymmetric oxygen” (O_a), and the “symmetric oxygen” (O_s). Correspondingly, it offers 2 sites for accommodating foreign ions: the zigzag intralayer site and the interlayer sites in between the bilayers.²⁷ It is reported that with low hydrogen incorporation, the one-dimensional (1D) zigzag intralayer sites of MoO_3 can be occupied first,³² accompanied by H bonding to O_a as $-\text{O}_a\text{H}$ ³³ to form early-phase HMBs (e.g., $\text{H}_{0.31}\text{MoO}_3$). After a certain limit, H bonds further with O_t to form $-\text{O}_t\text{H}_2$ ³³ and fills in interlayer sites that are favorable for 2D proton transport.³² Meanwhile, HMBs at high hydrogen levels are generated (e.g., $\text{H}_{1.68}\text{MoO}_3$). It is also reported that certain hydrogens in HMBs are hard to extract via chemical reactions,³¹ and the limit ($x = 0.85$) is postulated to be equivalent to the relative numbers of hydrogen in the intralayer sites.³² Interestingly, this “maximum occupancy” is very close to our experimental value ($x = 0.88$ in Figure 2A) of the first charge and explains the observed irreversible capacity loss. However, we also note that the characterization of higher-degree HMBs is less complete. There is controversy in the intralayer proton distribution in $\text{H}_{1.68}\text{MoO}_3$,^{32,34} where the average structure presents a large majority of protons as part of the $-\text{O}_t\text{H}_2$ group.³¹ The overall topotactic reaction is reported to proceed with minimal rearrangement, presumably due to similar structures being adopted.³³

The real situation is more complex. Additional reference structures are listed in Table S1. These HMBs references depict the average structure arrangements upon hydrogen loading. However, these are determined from chemically synthesized materials by dissociated (atomic) hydrogens,^{31,35,36} which are less likely to involve water. By contrast, protons are typically solvated in aqueous electrolytes, so electrochemical reactions may be different. Yan et al. first proposed that hydroniums inserted MoO_3 for charge compensation²² because (1) from the reported proton solvation energy of -269 kcal/mol (i.e., -11.66 eV, via probing gaseous $\text{Eu}(\text{H}_2\text{O})_n^{3+}$),³⁷ hydroniums are anticipated to require equivalent high energies to dehydrate (therefore, are stable in acids), (2) corresponding proton batteries function well at low temperatures—even below -40°C —which clearly differs from common deteriorations because of sluggish ion desolvation found in lithium-ion batteries, and (3) hydroniums are reported to present similar sizes (0.1 ± 0.01 nm³⁸, e.g., H_3O^+) as naked Na^+ , which can be hosted by MoO_3 . However, the key problems are (1) while protons can only covalently bond to individual oxygens forming $-\text{OH}_x$ groups, inserted alkali-metal cations would ionically coordinate with multiple adjacent oxygens and be held as polyhedrons [MO_n], and different structures result (Figure S12); (2) if assuming the potential insert of H_3O^+ equivalent to Na^+ , the lattice parameter changes of “ $(\text{H}_3\text{O})_x\text{MoO}_3$ ” (as Na_xMoO_3) upon ion intercalation would also be rather significant and distinct from those of H_xMoO_3 (Figure S13; Table S2); and (3) MoO_3 hydrates display quite different structures (Table S2) and the original bilayers of MoO_3 are possibly required to fracture if polarized H_2O enters (Figure S14³³). Therefore, an important question is again raised: what ions/molecules can enter MoO_3 during proton electrochemistry?

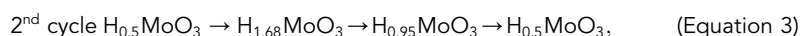
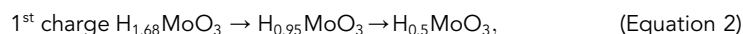
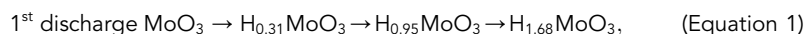
Operando synchrotron XRD data were collected to probe the structural evolution of the MoO_3 electrode in real time via a customized cell (Figure S15).³⁹ Figure 2C shows the XRD contour plot of cell 1 (at a rate of ~ 1.0 C, in a 4.4-M H_2SO_4 electrolyte). Characteristic potential plateaus are observed throughout, which are indicative of two-phase reactions in electrodes⁴⁰ and accord with the “battery-like” redox behavior observed in Figure 1. Refinements of structural models were carried out against the data of selected stages, labeled (i)–(x), to identify the phases present and determine their structural parameters (Table 1; Figure S16). A broad diffraction

Table 1. Structural Models and Refined Lattice Parameters against Data in Figure 2C

Stage	Dataset	Phase	Space Group	a/Å	b/Å	c/Å	Vol/Å ³
(i)	1	MoO ₃	<i>Pbnm</i>	3.9820(3)	13.9332(10)	3.7171(3)	206.231(42)
(ii)	8	H _{0.31} MoO ₃	<i>Cmcm</i>	3.9250(2)	14.0129(10)	3.7389(2)	205.643(27)
(iii)	45	H _{0.95} MoO ₃	<i>I12/m1</i>	13.9651(12)	3.7280(3)	7.6839(3)	399.364(50)
(iv)	57	H _{1.68} MoO ₃	<i>C12/m1</i>	13.5961(49)	4.0707(9)	3.7541(4)	207.375(41)
(v)	67	H _{0.95} MoO ₃	<i>I12/m1</i>	13.9684(12)	3.7192(5)	7.6927(5)	398.495(65)
(vi)	82	H _{0.5} MoO ₃	<i>Cmcm</i>	3.8082(1)	14.5915(8)	3.7372(2)	207.665(18)
(vii)	102	H _{1.68} MoO ₃	<i>C12/m1</i>	14.0221(6)	3.7791(1)	4.0648(1)	214.913(15)
(viii)	118	H _{1.68} MoO ₃	<i>C12/m1</i>	13.6184(18)	3.7848(4)	4.0697(2)	209.355(30)
(ix)	127	H _{0.95} MoO ₃	<i>I12/m1</i>	13.9207(3)	3.7363(4)	7.6660(9)	397.789(50)
(x)	143	H _{0.5} MoO ₃	<i>Cmcm</i>	3.7958(1)	14.5642(7)	3.7146(3)	206.852(13)

feature is observed in the $8.5^\circ < 2\theta < 15^\circ$ region, which can be attributed to contributions from the liquid electrolyte and Kapton tape used for sealing. The background profile prevents some expected reflections of the materials from being observed, and Pawley refinements of structural models were undertaken to determine lattice parameters.

Upon discharging and charging, a complex structural evolution is observed in which a series of two-phase and solid-solution reactions occur concurrently. This is also in clear contrast to the solid-solution evolution observed for pseudocapacitive tungsten oxide hydrates.^{19,41} Based on the modeling results of various stages (i.e., i–x), a reaction pathway for the major phases is suggested from a structural perspective:



Note that X-ray radiation provides low sensitivity to probing hydrogen atoms, and only a few well-determined structures are available for refinement. Therefore, the hydrogen content in the reference model is likely to differ from the true value of the electrode; but the lattice changes and peak position evolution are very informative and have led to these assignments. *Operando* experiments were also carried out to compare the effect of current and electrolyte concentration on the structural evolution. Two more identical cells were prepared and studied under different conditions: cell 2 (5.0 C, 4.4 M H₂SO₄; Figure S17) and cell 3 (1.0 C, 1.0 M H₂SO₄; Figure S18). In general, their structural behaviors appear consistent with those for cell 1. Similar structural evolution pathways were observed, and the same phases could be modeled against the diffraction data at similar electrochemical states (Figures S19 and S20; Tables S3 and S4). From the refined lattice parameters, similar changes for starting/ending states are found despite the conditions (Figure S21).

All of these findings suggest that atomic hydrogen-generated HMBs are adequate to model the XRD data as opposed to any molybdenum oxide hydrates. This is to say that regardless of the possible species (e.g., proton, hydronium) interacting with electrodes during electrochemistry, only the H⁺ is inserted into the crystal structure, and major reactions proceed within the parent MoO₃ and corresponding HMBs phases with no long-range ordered hydrates formed. This conclusion is also verified in multi-cycled electrodes (Figure S22).

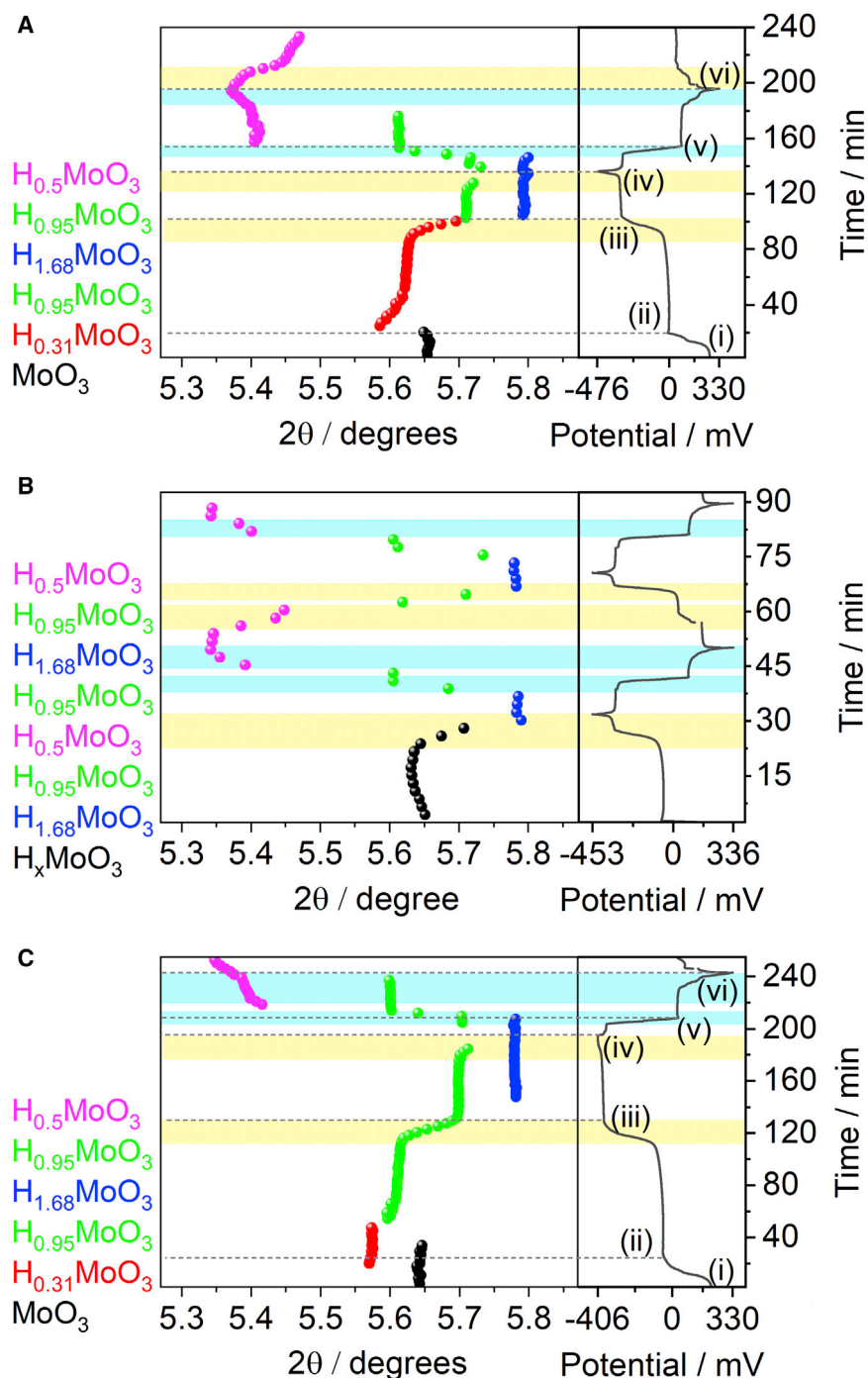


Figure 3. Dynamic Phase Transitions at Different Conditions

Single peak fitting results of (A) the first cycle of cell 1 (4.4 M H_2SO_4 , ~1 C), (B) the initial 2 cycles of cell 2 (4.4 M H_2SO_4 , ~5 C), and (C) the first cycle of cell 3 (1.0 M H_2SO_4 , ~1 C). The Roman numerals in (A) and (C) represent the same electrochemical states of different cells. The yellow/blue color highlights contraction/expansion of the interlayer spacing of slope regions.

Dynamic Phase Transitions at Different Conditions

To gain further structural insight, single peak fitting of key reflections was undertaken. According to Bragg's law, these reflections typically correlate with atomic

planes with the interlayer spacings of the crystal structure. Movements of these reflections show strong correlations with the electrochemical state of the cell throughout and provide evidence for some solid-solution behavior during operation, which could not easily be captured using the Pawley refinements. Figure 3A shows the low-angle reflection evolution of cell 1. Initially, the (002) reflection of MoO_3 is observed and remains generally constant with proton insertion. Reaching (ii), the original reflection sharply shifts to lower angles, indicating a solid solution reaction ($\text{H} < 0.31$), and a new reflection also forms corresponding to the $\text{H}_{0.31}\text{MoO}_3$ phase. From (ii) to (iii), the (020) reflection of $\text{H}_{0.31}\text{MoO}_3$ moves to higher angles corresponding to an overall contraction of the unit cell along the b axis. This coincides with a possible solid-solution reaction; meanwhile, H^+ is inserted into the structure to eventually form $\text{H}_{0.95}\text{MoO}_3$ at (iii) via a two-phase mechanism. From (iii) to (iv), it suggests a two-phase reaction to form $\text{H}_{1.68}\text{MoO}_3$ at (iv), and this reaction is then reversed to form $\text{H}_{0.95}\text{MoO}_3$ as the cell is recharged. From (iv) to (v) with further charge, a two-phase reaction to form $\text{H}_{0.5}\text{MoO}_3$ at (v) is observed; meanwhile, the (200) reflection of the $\text{H}_{0.95}\text{MoO}_3$ phase shifts to lower angles. The $\text{H}_{0.5}\text{MoO}_3$ phase then appears to exhibit solid-solution behavior until the (020) reflection reaches a minimum value at the charged state (vi). Symmetric interlayer-spacing changes repeatedly occur from this point for further cycles. Overall, the interlayer spacing contracts with proton insertion and expands upon proton extraction. This feature is reminiscent of decreased/enhanced repulsion of oxygen-oxygen from metal-oxygen layers in layered lithium transition metal oxides of lithium-ion batteries.^{42,43}

The structural responses to higher current rates are shown in Figure 3B. Due to the fast reaction rates, many details merge and overlap. However, similar evolution trends are still present: (1) after the initial regions, the same phases account for reflections of typical electrochemical states as in cell 1; (2) minimum/maximum interlayer spacings are also reached in discharged/charged states, respectively; (3) as color highlighted, the trends of interlayer-spacing change are similar to those of cell 1. The reflection positions of HMBs after the activation cycle display highly symmetric features, thus verifying the high reversibility of activated structures over successive cycles. All of these results suggest that the structure transformation associated with proton (de)insertion of MoO_3 would not be “frozen” by increased currents, as reported in anhydrous WO_3 .⁴¹ In other words, structure transformations are not likely the limiting process for the rate capability within certain ranges. In Figure 3C, cell 3 exhibits slightly different evolution at low angles, although overall it goes through similar changes during the electrochemical steps. The key difference appears to be the onset and the depth of certain reactions—for example, the amount of solid-solution reaction for a selected composition. The interlayer spacing keeps expanding after (vi) instead of contracting back as with the other 2 cells. The differences in dilute electrolytes are essentially related to water influences.

Overall, the important findings are that HMBs alone fit the real-time data during functioning rather than with any hydrates; the structure transformation triggered by bulk proton diffusion, although complicated, is still fast, therefore, it supports the good rate capability. The sequence of phase transitions in electrochemistry is similar to the chemical hydrogen loading, but these are likely to show some differences as the hydrogen content in refined models could be different from the real value. It should again be noted that the hydrogen content in the modeled HMBs relies heavily on the lattice parameters, as detecting atomic hydrogen with XRD is very difficult in a pure powder sample, let alone a complex electrochemical cell. This means that minor changes in hydrogen content are difficult to probe.

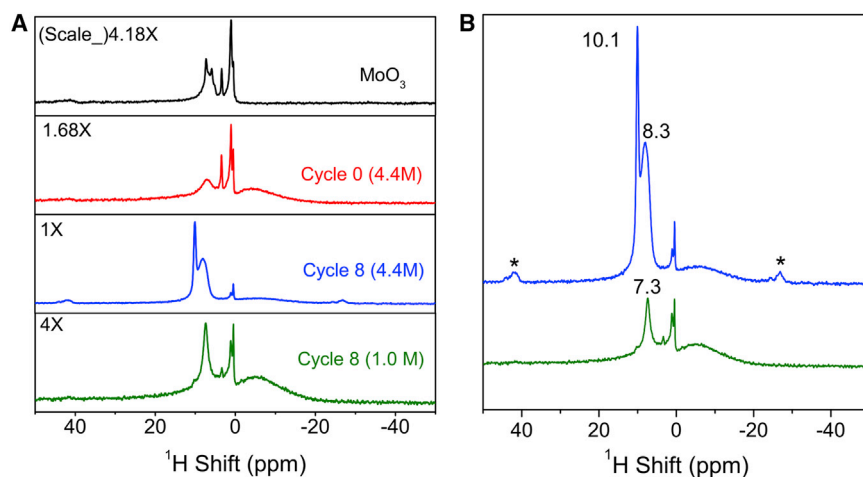


Figure 4. ^1H Solid-State NMR Detection of Water on Electrode Surfaces

(A) The spectra of original MoO_3 powders and processed electrode samples. Cycle 0 represents electrode soaked into electrolyte for the same amount of time as the cycling electrodes before the following treatments. The text at the top left of each panel indicates the scale factor of intensities for highlighting detailed features.

(B) The specific spectra of the electrode samples cycled 8 times in 4.4 M electrolyte (blue) and 1.0 M electrolyte (olive). The asterisks represent the spinning sidebands of the isotropic peaks.

Detection of Water on Electrode Surfaces

Interestingly, although water and hydronium do not insert into bulk crystal structures, there are many hints in the previous sections to reflect water effects. This raises questions of (1) is any form of water present in materials and where? (2) if at the surface, how would the surface interactions proceed? and (3) what influences would it have on battery function?

To investigate the potential existence of water in the electrodes, *ex situ* ^1H solid-state NMR (ssNMR) experiments were performed. ^1H NMR has been reported for investigating the structure and dynamics of hydrogen in pure HMBs but with low resolution.^{30,32,44} Here, high magnetic field and fast magic angle spinning (MAS) are used to gain signals of distinct hydrogen environments of cycled HMB electrodes. As shown in Figure 4A, the spectrum of neat MoO_3 presents relatively weak ^1H NMR signals, attributed to residual impurities and background signal from the probe head. The sample only soaked in electrolytes without electrochemistry was also tested, designated as "cycle 0." The corresponding spectrum exhibits similar ^1H signals as the MoO_3 , but with an additional signal at ~ 5 ppm, which indicates adsorbed water species from the electrolytes. These spectra represent the baselines for comparison of the cycled electrode samples.

For electrodes cycled in 4.4 M electrolytes (Figure 4B), there are strong peaks at 10.1 and 8.3 ppm that are of direct relevance. The narrow 10.1 ppm signal is assigned to strongly hydrogen-bonded OH species within the HMB structures. By contrast, the broader signal at 8.3 ppm is attributed to H_2O species within the HMB crystal. The OH and H_2O species are discriminated on the basis of their spinning sidebands (ssbs), where the signal at 10.1 ppm yields very weak ssbs, while the signal at 8.3 ppm yields relatively more intense ssbs (even at the fast 24 kHz MAS) due to the dipolar coupling between the two geminal protons in H_2O . Although different from the nominal 4.7 ppm signal for bulk crystal water, the high chemical shift of 8.3 ppm confirms that the H_2O species in the HMB are also strongly hydrogen bonded. Considering the diffraction results, the electrochemically absorbed water is likely to be confined at the surfaces of materials with a

disordered environment, which is consistent with the observed broad nature of the 8.3 ppm signal. Repeat measurements on another electrode cycled in 1.0 M H_2SO_4 were also conducted. The dilute electrolyte is expected to involve more solvent-water interactions with the electrodes. However, only a weak threshold signal at ~ 10 ppm was observed. A strong hydrogen-bonding signal from H_2O species at 7.3 ppm is still present, but it is much narrower than the corresponding 8.3 ppm signal for the 4.4 M H_2SO_4 counterpart. Demonstrated by the absence of dipolar ssbs, H_2O species in the electrode of the dilute group turned out to be surface adsorbed with a relatively higher degree of motion.³² These results again suggest that the solvation water plays a role in the electrochemical proton storage, but predominantly at the surface level.

Operando Mass Change

EQCM tests were performed to monitor the *operando* mass change and to identify the active charge carriers, as well as their interactions with the MoO_3 electrode. In aqueous solutions, protons are mostly bonded to water to form hydroniums, which can influence surface electrochemistry (Figures S8–S10) and induce overall mass changes of electrodes (Figure S23; Note S3). Because the proton (H^+) itself has negligible weight, the mass change can be mostly attributed to the concurrent interactions of hydronium/water with electrodes. Figure 5A shows the CVs and corresponding mass-potential (m - E) curves of the first cycle. The CVs correlate well with the charge-discharge curves in Figure 2A. In the m - E curve, the mass increases through the C1' and C2' processes as a result of hydronium adsorption onto the MoO_3 surface. These are followed by naked proton insertion into bulk lattices, which triggers the initial irreversible phase transition from orthorhombic MoO_3 to monoclinic HMBs and then the subsequent formation of higher degree HMBs through C3, as disclosed above. In mass, an increase is first observed for C3 but is immediately followed by a loss, which can be attributed to the opposite transport of adsorbed water.^{45,46} A similar feature has also been observed in hydrated tungsten oxides.¹⁹ These suggest a vehicle-type proton conduction mechanism,¹¹ in which protons are transported by the carrier vehicle (e.g., hydronium, $\text{H}(\text{H}_2\text{O})_n^+$) with the empty vehicle (e.g., H_2O) moving in the opposite direction. Interestingly, it is noted that the C3 process shows faster kinetics than C2', as reflected by the much-reduced polarization (Figures 5B and S3), which indicates that the concurrent water movements can significantly facilitate the charge transfer. On the reverse scan, the electrode mass increases with proton deintercalation through the A3, A2, and A1 processes, again due to the opposite transport of water. At the end of the first cycle, a significant increase in the overall electrode mass (i.e., water accumulation) is found, which agrees well with the *ex situ* weighing results (Figure S23).

The CVs of the second cycle (Figure 5C) also accord with the charge-discharge curves in Figure 2A. Compared to the first cycle, the C1 and C2 peaks start to merge together and become more symmetric with the anodic A1/A2 peaks, which reflect more reversible structure transformations within HMBs after the first cycle. The mass increase is initially found for each of the C1, C2, and C3 processes due to hydronium adsorption and is quickly followed by a mass drop due to the opposite transport of water. In the reverse scan, the mass increase is again observed for A3 and A2/A1 processes due to the same reason. Overall, water content in the electrodes keeps increasing in the second cycle, but to a reduced extent, as reflected by the narrowed gap between the start point and endpoint. In the following cycles (Figure 5D), the CVs remain similar to the second cycle, but mass changes become more and more reversible, and eventually the overall mass stays generally unchanged after each cycle. These mass evolutions accord with the good electrochemical reversibility after activation. Otherwise, the subtle mass gap may still intermittently occur after a whole scan (Figure S24); this accounts for the slow material dissolution observed (Figures 1E, S7, and S23), which are probably physical

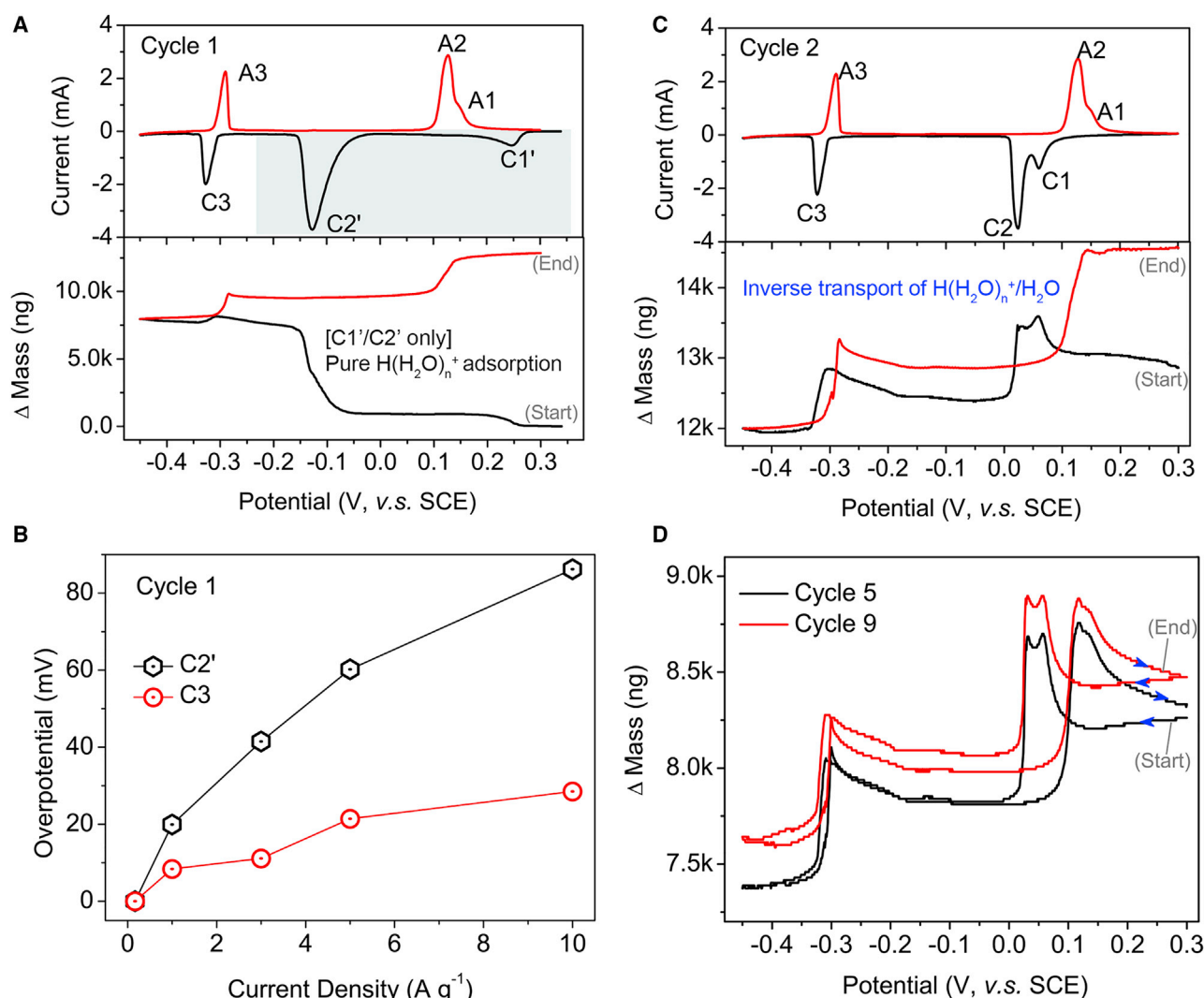


Figure 5. Operando Mass Changes during Electrochemistry

(A) EQCM results for MoO₃ anodes of CV (top, at 3 mV/s) and corresponding mass response (bottom) in the first cycle. In the gray colored region (C1'/C2'), pure hydronium adsorptions occur, since C3 of cycle 1 inverse hydronium/water transportations are the dominant surface interactions for the following electrochemistry.

(B) The overpotentials with respect to the increased current density of C2' and C3 redox during the first galvanostatic charge-discharge. Overpotentials are calculated based on results in Figure S3, with the values of 1/6 A g⁻¹ current density as the standard.

(C) The corresponding EQCM results in the second cycle of the same electrode in (A).

(D) The mass potential curves of cycles 5 and 9 of another electrode in Figure S24, indicating reversible overall mass changes after the initial surface hydrations.

processes but are caused by the hydronium/water activities during electrochemistry. The dissolution is identified as the major reason for cycling performance decay, given the consistent interface resistance and bulk proton diffusion coefficient after activation (from impedance results, Figure S25). Thermogravimetric analysis (TGA, Figure S26; Note S4) of the cycled electrodes further verifies the interfacial water accumulation and material dissolution; and the enhanced cycling stability at a lowered operation temperature (5°C, Figure S27) again reflects the influence of water activities.

The above EQCM results reveal that hydronium is the active charge carrier at electrode-electrolyte interfaces. Overall, the protonation topochemistry of α -MoO₃ can be

summarized to proceed via two key steps—hydronium adsorption on the surface and naked proton insertion into the bulk lattices—that unify the referred disputes on the actual ion-charge carrier.^{20,22} The hydronium interactions with the MoO₃ surface form a hydrated layer, where proton transport proceeds via a vehicle mechanism with the opposite transport of water and much reduced polarization. This is in accordance with the solid-state NMR results, which show that water is detected but only on the surface. Surface hydrations lead to a net mass increase mainly in the first few cycles until water adsorption/desorption reaches equilibrium, then mass change becomes reversible for the following charge/discharge processes. Across the hydrated layer, naked protons alone can insert deeply into the lattices and trigger the protonation of bulk MoO₃, as confirmed by *operando* XRD studies. Inside the lattices of MoO₃, proton transport is presumed to occur via a lattice-oxygen-mediated hopping, as interpreted in a similar class of materials, tungsten oxide hydrates.⁴⁷ The insertion/extraction of naked protons into/from bulk MoO₃ accounts for most of the capacity observed, while hydronium/water plays important roles on the interfacial reactions and electrode stability.

In summary, proton transport and storage mechanisms are studied in α -MoO₃ for proton batteries. It is found multiple ions/species interact with electrodes at different stages, which benefits kinetics. *Operando* synchrotron XRD reveals that proton rather than hydronium inserts into the bulk materials, and the topotactic reactions occur only between MoO₃ and corresponding HMBs. Surprisingly, the phase transitions, although complicated, are found not to limit the reaction kinetics, thus facilitating the high rate-capabilities. ssNMR detects water in the cycled electrodes, which is confined at the surfaces of the materials. EQCM measurements confirm the surface hydrations of MoO₃ and disclose hydronium is instead the charge carrier for proton transport at the hydrated surface, which are accompanied by water adsorption/expelling and enhanced charge transfer. Otherwise, slow material dissolutions are induced by the water activities, instead of acidic corrosions, and account for the major reason for cycling performance decay. It is our hope that this study can inspire further research and development in searching for and exploiting novel proton storage materials and systems, which lay the foundation for a promising high rate and capacity energy storage.

EXPERIMENTAL PROCEDURES

Resource Availability

Lead Contact

Further information and requests for resources should be directed to the Lead Contact, Prof. Chuan Zhao (chuan.zhao@unsw.edu.au).

Materials Availability

This study did not generate new unique materials.

Data and Code Availability

The authors declare that data supporting the findings of this study are available within the article and the [Supplemental Information](#). All of the other data are available from the Lead Contact upon reasonable request.

Electrochemical Measurement

The electrochemical properties of α -MoO₃ (Sigma-Aldrich) were studied via the three-electrode system. Typically, the active materials worked as the working electrode (WE), graphite worked as the counter electrode (CE), and a saturated calomel electrode (SCE; E = 0.241 V, as compared to standard hydrogen electrode [SHE]) was used as the reference electrode (RE), and the cells were tested in sulfuric acid with different concentrations. To prepare the MoO₃ electrodes, 70 wt% active materials, 20 wt% acetylene

black (conducting agents), and 10 wt% polyvinylidene fluoride (PVDF, binders), were ground, mixed then dispersed in *N*-methyl-2-pyrrolidone (NMP) under stirring for several hours. Afterward, the obtained slurry was drop-cast onto titanium plates (working area $\sim 1 \times 1 \text{ cm}^2$) and then dried at 60°C overnight. These worked as the default electrodes for most of our electrochemical evaluations. Otherwise, carbon fiber paper (CFP) substrates were also processed the same way for additional cycling. The active-mass loading was 3–5 mg/cm^2 for titanium substrates (unless otherwise noted), and on average, $\sim 6 \text{ mg}/\text{cm}^2$ for CFP substrates. Galvanostatic charge/discharge measurements were conducted via a battery testing system (LAND CT2001A); CV tests were performed on a CHI 440C potentiostat.

Physical Characterization

ssNMR spectra were obtained on a Bruker Advance III 700 MHz Solid-State NMR spectrometer, operating at 700 MHz for the ^1H nucleus, spinning at MAS frequencies of 24 kHz. Electrode samples (prepared by using 10 wt% polytetrafluoroethylene [PTFE] as binders, with 10 wt% acetylene black and 80 wt% active materials) were electrochemically processed, water washed thoroughly, and then dried at 70°C under air overnight. After that, the samples were scratched off the substrates and filled ($\sim 10 \text{ mg}$) into 3.2-mm zirconia rotors, fitted with Vespel caps. ^1H chemical shifts were externally referenced to adamantane at 1.8 ppm. All of the NMR experiments were performed at ambient temperature.

Operando Structural Analyses

Operando synchrotron XRD experiments were performed on the Powder Diffraction beamline at the Australian Synchrotron⁴⁸ at a wavelength of $\lambda = 0.686867(31)$, which was determined using the National Institute of Standards and Technology (NIST) 660b LaB_6 standard reference material.⁴⁹ Data were collected in the range of $2^\circ < 2\theta < 76^\circ$ with a collection time of 60 s per position (totally $\sim 140 \text{ s}$ between each pattern, including the detector position movement), in transmission geometry throughout cycling. A customized cell³⁹ in a three-electrode configuration was adopted for synchrotron tests during real time electrochemistry (see Figure S15 and the Supplemental Experimental Procedures for more details). The structure diagrams were drawn using VESTA software.

EQCM Measurement

EQCM measurements were conducted at ambient temperatures via the CHI 440C potentiostat with an oscillator. The cell also adopted a three-electrode system, using the gold-coated quartz to load materials as WE, platinum wire as CE, and a SCE as RE. To prepare WE, 70 wt% active materials, 20 wt% acetylene black, and 10 wt% PVDF were dispersed in NMP to form a relatively dilute slurry. The slurry was dripped on the gold surface of the quartz crystal, which was dried under 60°C overnight. The shift of the quartz resonance frequency (Δf) was converted into the mass change (Δm) of the electrode-coated quartz crystal using Sauerbrey's equation⁵⁰:

$$\Delta m = -\frac{A\sqrt{\mu \cdot \rho}}{2f_0^2} \cdot \Delta f = -C_f \cdot \Delta f$$

where ρ is the density of quartz, μ is the shear modulus of quartz (2.947 GPa), f_0 is the fundamental resonant frequency (7.995 MHz), A is the area of the gold disk coated onto the crystal ($\sim 0.204 \text{ cm}^2$), and C_f is the sensitivity factor, which is practically calculated to be 1.41 ng/Hz .

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.xcrp.2020.100225>.

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AUTHOR CONTRIBUTIONS

C.Z., H.G. conceived the project. H.G. performed the experiments and data analysis. H.G., D.G. conducted the synchrotron characterizations and D.G., N.S. analyzed the data. Z.S. contributed to the electrochemical/X-ray photoelectron spectroscopy (XPS) tests. A.R. recorded and analyzed the ssNMR results. W.R. and C.Z. supervised the research. All of the authors contributed to the discussion and the writing of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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